# Project 1

# Data Analysis on Railway Gauges

## Objective:

The main objective of this project is to understand the trend between the usage of gauges in railways. We are going to understand which gauge either narrow, broad or meter gauge is more prominently used in railway services. We are analyzing historical data.

## Introduction:

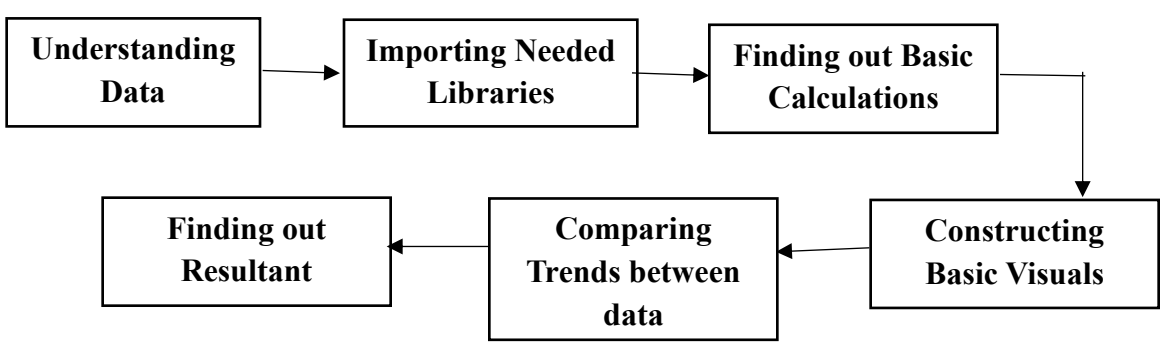
In this project, we are performing basic data analytics using the historical data. We are going to understand about *"what has happened before"* in this analysis. The data analytics has been performed using pandas, NumPy and matplotlib libraries in order to understand the statistical behaviors of the data and understand the trend happening in between the data. Now, let's analyze the dataset. The below are the requirements:

- IDLE
- Pandas
- NumPy
- Matplotlib

## Project Workflow:

The basic workflow of this project discusses about the below things.

1. Firstly, we are going to analyze and understand what the data actually saying. Here it just contains the 4 columns with 50 rows. It had three kinds of gauges values with their respective years. The data, details about the usage of gauges in several years.
2. Then, we can perform *historical data analysis* so that it answers the questions like:
    - Which gauge is used and in more demand ?
    - Finding in which year the gauge values are more ?
    - Analyzing the trend between gauges by visualizing them in line chart.
    - Finding usage of particular gauge and understanding its prominence.
3. To perform overall analysis, we import dataset and then do the following steps.



4. We perform analysis in the above flow in order to make a clear understanding on the analysis we do.

## Steps for Constructing Analysis:

### *Step – 1:*

Here, in this step we are going to import dataset into a pandas data frame as below.

```
#importing modules
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

#Reading dataset from csv file
data = pd.read_csv("C:/Users/Admin/Documents/pyCodes/data_Analytics/project_1/railway_gauges.csv")
```

Then, we are going to display the first 5 rows and the last 5 rows of the dataset in order to understand the structure of the data we have.

```
#displaying first 5 and last 5 rows
print("First Five Rows: ")
print(data.head())
print("----------------------------------------------------------------------")

print("Last Five Rows")
print(data.tail())
print("----------------------------------------------------------------------")
```

So we have used head() and tail() functions in order to display the data.

- head() is a function used to display the default first 5 rows of the dataset.
- tail() is a function used to display the bottom 5 rows of the dataset.

We get result as below snippet.

```
== RESTART: C:\Users\Admin\Documents\pyCodes\data_Analytics\project_1\main.py ==
First Five Rows:
      Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
0  1964-65         3521         2891           464   6876
1  1965-66         3603         2913           470   6986
2  1966-67         3615         2911           481   7007
3  1967-68         3601         2920           506   7027
4  1968-69         3603         2921           508   7032
----------------------------------------------------------------------
Last Five Rows
       Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
44  2008-09         6185          642           203   7030
45  2009-10         6320          543           220   7083
46  2010-11         6419          494           220   7133
47  2011-12         6494          432           220   7146
48  2012-13         6577          379           216   7172
----------------------------------------------------------------------
```

### *Step – 2:*

Now to know the basic information about the data we use info() method which will give us basic details like data type of the dataset, size of the data etc.

```
#Finding out basic information
print(data.info())
print("-----------------------------------------------------------------")
```

So after running the above code we get the output as,

```
<class 'pandas.DataFrame'>
RangeIndex: 49 entries, 0 to 48
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Year          49 non-null     str
 1   Broad Gauge   49 non-null     int64
 2   Metre Gauge   49 non-null     int64
 3   Narrow Gauge  49 non-null     int64
 4   Total         49 non-null     int64
dtypes: int64(4), str(1)
memory usage: 2.0 KB
None
-----------------------------------------------------------------
```

***Step – 3:***

So now we are using describe() function to know the statistical information of the dataset. It is a default function which gives mean, count, minimum, maximum, standard deviation, quartile deviation information.

```
#Statistical Information regarding dataset
print("Statistical Analysis: ")
print(data.describe())
print("-----------------------------------------------------------------")
```

So the statistical values what we have got after executing the above code is:

```
-----------------------------------------------------------------
Statistical Analysis:
       Broad Gauge  Metre Gauge  Narrow Gauge        Total
count    49.000000    49.000000     49.000000    49.000000
mean   4572.775510  2077.510204    386.795918  7037.081633
std     946.656887   847.318863    106.809991    73.805724
min    3521.000000   379.000000    191.000000  6853.000000
25%    3767.000000  1318.000000    284.000000  7017.000000
50%    4163.000000  2478.000000    442.000000  7056.000000
75%    5261.000000  2786.000000    468.000000  7084.000000
max    6577.000000  2921.000000    508.000000  7172.000000
-----------------------------------------------------------------
```

***Step – 4:***

Now, we are going to find out about which gauge values are more in order to understand the maximum values of every gauge. If we find which gauge values are higher, then we can understand its importance usage in railways.

```
#1. Finding in which year the gauge values are more ?
broad_gauge_max = data.loc[data['Broad Gauge'].idxmax(),'Year']
print(f"Broad Gauge Maximum value in year: {broad_gauge_max}")

narrow_gauge_max = data.loc[data['Narrow Gauge'].idxmax(),'Year']
print(f"Narrow Gauge Maximum value in year : {narrow_gauge_max}")

metre_gauge_max = data.loc[data['Metre Gauge'].idxmax(), 'Year']
print(f"Metre Gauge Maximum value in year: {metre_gauge_max}")
print("-------------------------------------------------------------------")
```

So here I have calculated the maximum values of every gauge just by using idxmax() function. This function helps to return the index labels where the first occurrence of the maximum value is located.

```
-------------------------------------------------------------------
Broad Gauge Maximum value in year: 2012-13
Narrow Gauge Maximum value in year : 1968-69
Metre Gauge Maximum value in year: 1968-69
-------------------------------------------------------------------
```

## ***Step – 5:***

Now in this step we will calculate the overall maximum usage of the gauges from the given dataset.

```
#2. Which gauge is having more demand overall?
broad_gauge = data['Broad Gauge'].sum()
narrow_gauge = data['Narrow Gauge'].sum()
metre_gauge = data['Metre Gauge'].sum()
print("Broad Gauge total usage in past years is: ",broad_gauge)
print("Narrow Gauge total usage in past years is: ",narrow_gauge)
print("Metre Gauge total usage in past years is: ",metre_gauge)
print("-------------------------------------------------------------------")
```

So here the maximum values of gauges are displayed as below snippet.

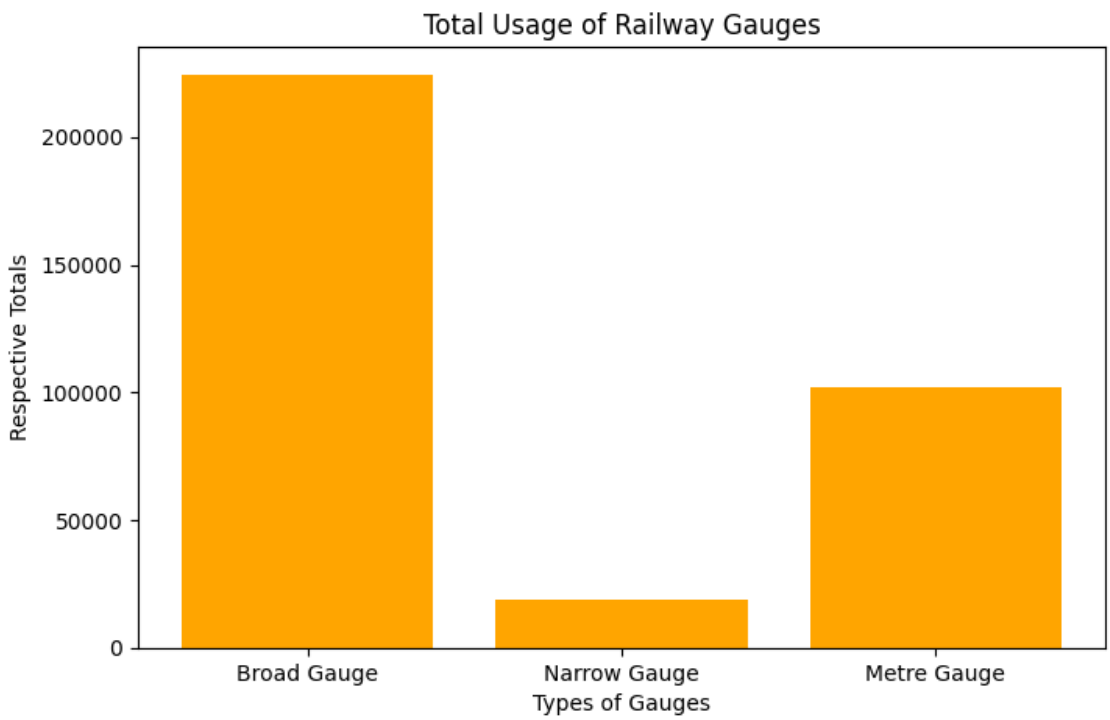
```
-------------------------------------------------------------------
Broad Gauge total usage in past years is:  224066
Narrow Gauge total usage in past years is:  18953
Metre Gauge total usage in past years is:  101798
-------------------------------------------------------------------
```

## ***Step – 6:***

Now, we are plotting the maximum values of each gauge by using a bar plot. We can visually see the higher demanded gauge by which we can get some conclusion regarding the higher gauge usage.  We have coded for bar plot as below.

```
#Plotting them to view it visually
gauges = ["Broad Gauge","Narrow Gauge","Metre Gauge"]
values = [broad_gauge, narrow_gauge, metre_gauge]
plt.figure(figsize = (8,5))
plt.bar(gauges, values, color = 'orange')
plt.xlabel("Types of Gauges")
plt.ylabel("Respective Totals")
plt.title("Total Usage of Railway Gauges")
plt.show()
```

Now when we execute this code, it takes values from gauges and values variable and adds them to the bar chart and displays as below



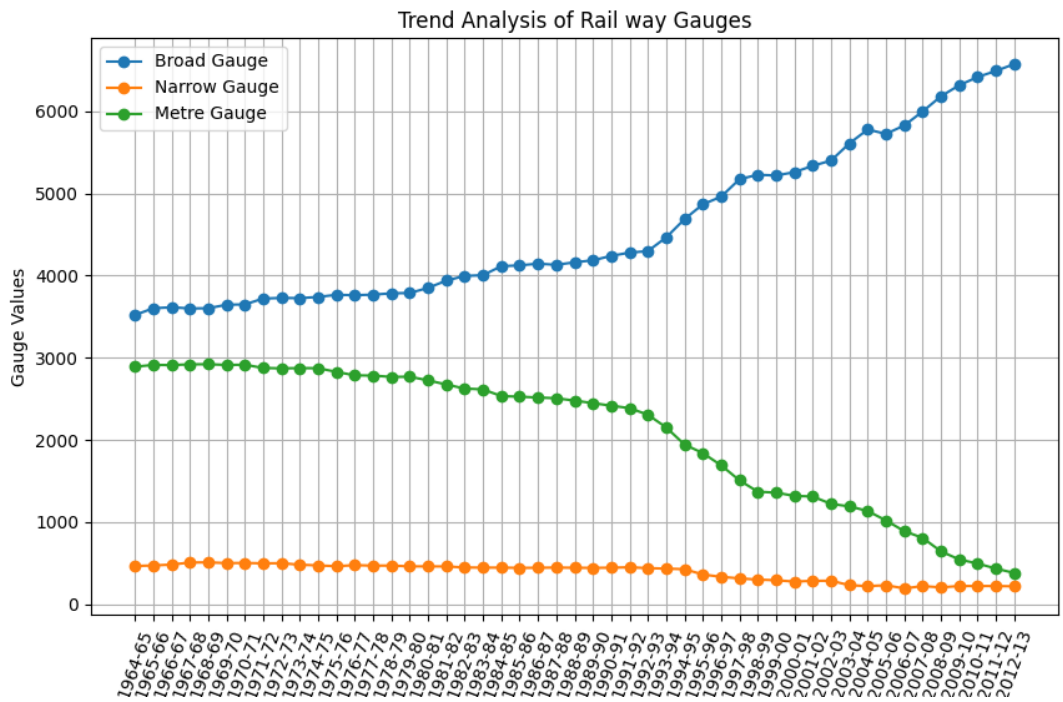

## *Step – 7:*

Finally, in this step, we are going to analyze the overall trends between different kinds of gauges as below.

We have plotted Broad Gauge, Metre Gauge and Narrow Gauge along with Years. So that we can see the incline and the decline of the each gauge along its time period. Here we can see three kinds of line plot representing each gauge along with plotting points. This is our final visual to conclude our analysis

```
#3. Analysing the trends between gauge to gauge just by plotting them using
#years and those respective values
plt.figure(figsize = (10,6))
plt.plot(data['Year'],data['Broad Gauge'],marker = 'o',label = 'Broad Gauge')
plt.plot(data['Year'],data['Narrow Gauge'],marker = 'o',label = 'Narrow Gauge')
plt.plot(data['Year'],data['Metre Gauge'],marker = 'o',label = 'Metre Gauge')
plt.xlabel("Total Years")
plt.ylabel("Gauge Values")
plt.xticks(rotation = 70)
plt.title("Trend Analysis of Rail way Gauges")
plt.legend()
plt.grid()
plt.show()
```

Now, let us see how our visual beautifully ended showing all these numbers.




## *Step – 8:*

So, in order to understand which gauge is most useful in the railway services, we have calculated the maximum of three gauges which directly represents the most used gauge.

The below is the code we have written to find out the most used gauge in railways services.

```
#Finding out most used gauges
totals = data[['Broad Gauge','Metre Gauge','Narrow Gauge']].sum()
maximum_gauge = totals.idxmax()
print("Most Used Gauge in all the three are: ",maximum_gauge)
```

The below is the output

```
-----------------------------------------------------------------------
Most Used Gauge in all the three are:  Broad Gauge
```

## Conclusion:

The analysis of railway gauge data with multiple years gives us clear understanding about kinds of gauges, evolution and utilization of different gauge types. By viewing the dataset, I have observed that each gauge reached its peak usage in different years, but mostly there is drastic difference between broad gauge and narrow gauge. There is so much changes existing in infrastructure development over time.

The overall usage comparison revealed that Broad Gauge has the highest total value, making it the most dominant gauge type in the railway network. Trend analysis further supports this finding, as Broad Gauge shows a consistent increase over time while Metre Gauge is declining in pattern.

The visualizations, including bar charts and line plots, effectively highlight these patterns and provide a clear understanding of gauge distribution and trends. It helped me how a data driven approaches can be uncovered and can get meaningful insights about infrastructure development.

## Scope of Developments:

As we have analyzed only historical or past data, it is only helpful for descriptive analysis and we got insights regarding past trends and patterns in railway gauge usage. In order to proceed with this analysis, it stood as a starter to get basic understanding of the dataset and its variables. But we can furtherly add predictive analysis module just by integrating additional datasets such as traffic, volume, route information, more comprehensive insights can be derived.